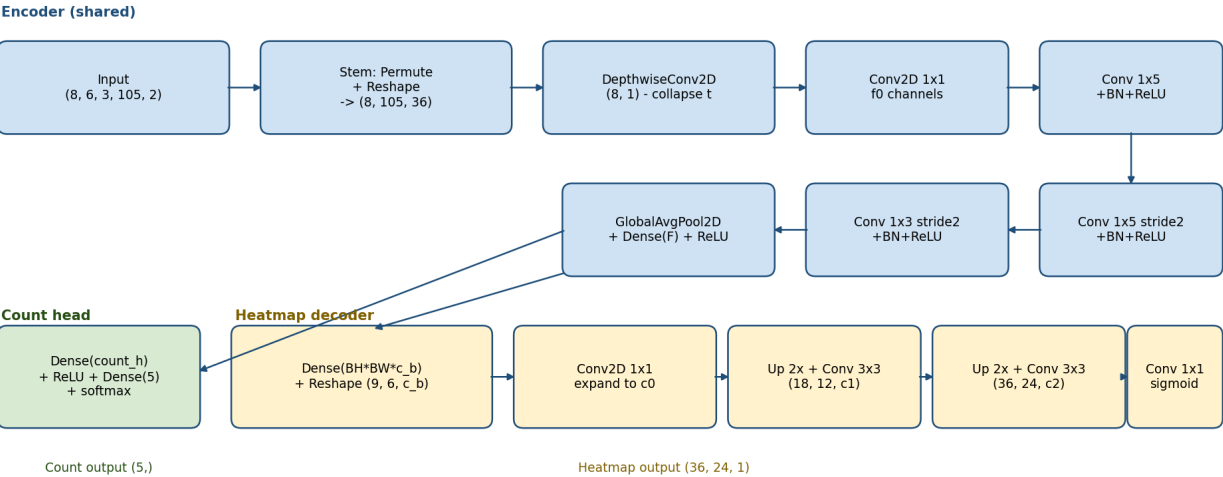


Model Architecture Briefing - Multi-Person UWB Localization

Generated 2026-05-01 01:10 - 3 size variants - Hybrid heatmap + count head - All TFLM-supported ops - INT8 quantized; tested on untrained weights for size verification

1. Architecture (shared across variants)

All three variants share the same encoder + decoder + count head structure; they differ only in the channel widths of each stage. The encoder maps the (8, 6, 3, 105, 2) stacked input to a flat feature vector via a depthwise temporal collapse followed by four range-direction convolutions. The heatmap decoder expands a (9, 6) bottleneck back to the (36, 24) room grid via two 2x upsampling stages. The count head is a small MLP on the same bottleneck features.



2. Variant configurations

The three variants are designed to span the size budget. Channel widths increase monotonically; the decoder bottleneck channel count c_b is small to keep the Dense projection from dominating the parameter budget.

Variant	Encoder channels	Feature width	c_b	Decoder ch (c_0, c_1, c_2)	Count hidden
small	32, 48, 64, 96	96	8	48, 32, 16	48
medium	48, 80, 112, 160	160	12	80, 56, 28	80
large	64, 112, 160, 224	224	16	112, 80, 40	128

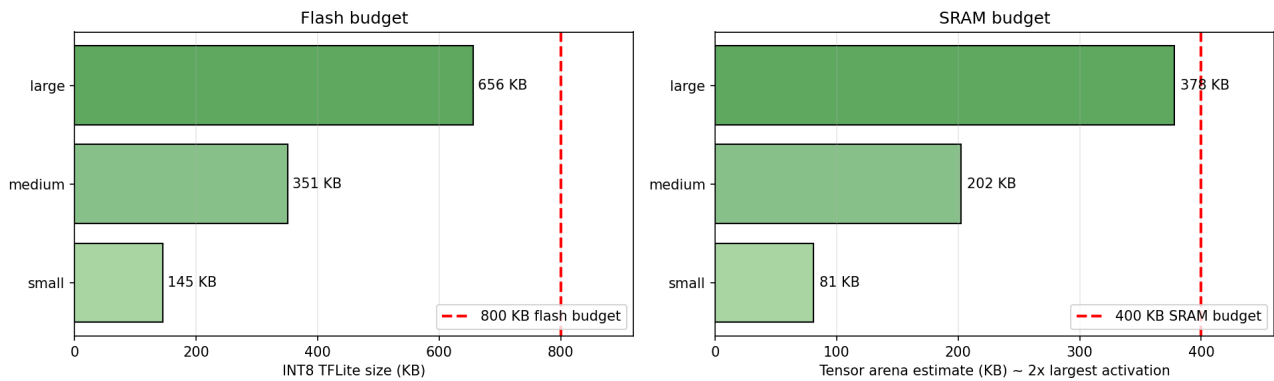
3. Parameter counts and TFLite binary sizes

TFLite sizes measured from real conversions (concrete-function path, fixed batch=1, BN moving stats baked as constants by save+reload). INT8 quantization is full post-training quantization with random calibration data; trained models will use real preprocessed CIR snippets and produce the same binary size.

Variant	Params	Float32 (raw)	Float32 TFLite	INT8 TFLite	Compression vs raw
small	119,350	466.2 KB	471.5 KB	145.0 KB	3.21x
medium	320,970	1253.8 KB	1255.3 KB	350.7 KB	3.57x
large	624,382	2439.0 KB	2436.5 KB	655.8 KB	3.72x

4. ESP32-S3 budget compliance

Variant	INT8 size	Flash limit	Flash OK?	Largest tensor	Arena est.	SRAM limit	SRAM OK?
small	145.0 KB	800 KB	Y	40.5 KB	81.0 KB	400 KB	Y
medium	350.7 KB	800 KB	Y	101.2 KB	202.5 KB	400 KB	Y
large	655.8 KB	800 KB	Y	189.0 KB	378.0 KB	400 KB	Y



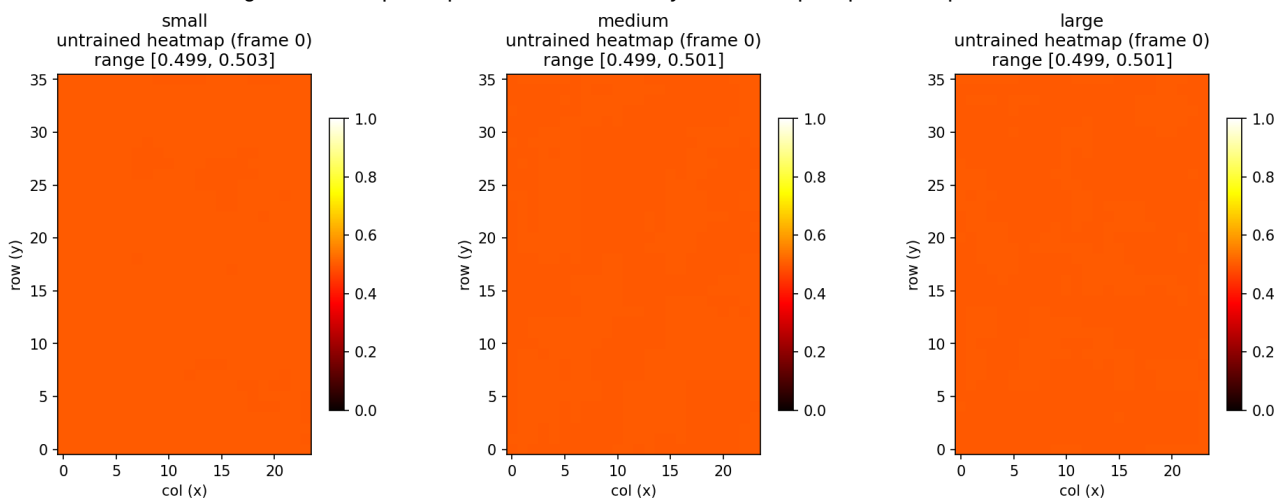
5. TFLite operation set

The INT8 graph contains only operations on the TFLM supported list. No LSTM/GRU/RNN, no Conv2DTranspose. Counts shown for the medium variant; the small and large variants use an identical op set.

Op	Count	TFLM supported?
CONV_2D	8	Yes
DELEGATE	5	Yes
DEPTHWISE_CONV_2D	1	Yes
FULLY_CONNECTED	4	Yes
LOGISTIC	1	Yes
MEAN	1	Yes
RESHAPE	8	Yes
SOFTMAX	1	Yes
TILE	4	Yes
TRANSPOSE	1	Yes

6. Untrained output sanity check

Untrained models produce a near-uniform sigmoid heatmap around 0.5 -- the network has not yet learned to distinguish presence from absence. After training, the heatmap collapses toward zero everywhere except at predicted person cells.



7. Conversion recipe

TFLite INT8 conversion of a Keras model with BatchNormalization is fragile. Several common paths fail: `dynamic_batch` with `from_keras_model()` falls back to TF Conv2D; `from_concrete_functions()` raises `READ_VARIABLE` errors during INT8 calibration; `from_saved_model()` likewise fails when BN moving stats are not folded. The reliable recipe used here, encapsulated in `convert_to_tflite()`, is:

1. Build a fresh inference-only model with `batch_size=1`.
2. Copy weights from the trained dynamic-batch model.
3. Run a dummy forward pass with `training=False`.
4. Save as `.keras`, reload via `load_model()` (this bakes BN moving stats as constants).
5. Convert with `TFLiteConverter.from_keras_model()` and INT8 representative dataset.